

An Analysis of Autonomous vehicle

Autonomous vehicle

Autonomous vehicle -- Part 1

Total incidents The US National Highway Traffic Safety Administration began mandating incident reports from autonomous vehicle companies in June 2021. Some reports cite incidents from as early as August 2019, with current data available through June 17, 2024. There have been a total of 3,979 autonomous vehicle incidents (both ADS and ADAS) reported during this timeframe. 2,146 of those incidents (53.9%) involved Tesla vehicles.

Autonomous vehicle -- Part 2

Privacy Some ACs require an internet connection to function, opening the possibility that a hacker might gain access to private information such as destinations, routes, camera recordings, media preferences, and/or behavioral patterns, although this is true of any internet-connected device.

Autonomous vehicle -- Part 3

Definitions Organizations such as the global standards body SAE International (SAE) have proposed terminology to describe technical capabilities. However, most terms have no standard definition and are employed variously by vendors and others. Proposals to adopt aviation automation terminology for cars has also not prevailed. The first consideration is the operational design domain (ODD). The concept presumes that automated systems have limitations. Relating system function to the ODD it supports is important for developers and regulators to establish and communicate safe operating conditions. Systems should operate within those limitations. Some systems recognize the ODD and modify their behavior accordingly. For example, an autonomous car might recognize that traffic is heavy and disable its automated lane change feature. Vendors have taken a variety of approaches to the self-driving problem. Tesla's approach is to allow their "full self-driving" (FSD) system to be used in all ODDs as a Level 2 (hands/on, eyes/on) ADAS. Waymo picked specific ODDs (city streets in Phoenix and San Francisco) for their Level 5 robotaxi service. Mercedes Benz offers Level 3 service in Las Vegas in highway traffic jams at speeds up to 40 miles per hour (64 km/h). Mobileye's SuperVision system offers hands-off/eyes-on driving on all road types at speeds up to 130 km/h (81 mph). GM's hands-free Super Cruise operates on specific roads in specific conditions, stopping or returning control to the driver when ODD changes. In 2024 the company announced plans to expand road coverage from 400,000 miles to 750,000 miles (1,210,000 km). Ford's BlueCruise hands-off system operates on 130,000 miles (210,000 km) of US divided highways. Names such as AutoDrive, PilotAssist, "Full-Self Driving" or DrivePilot are used even though the products offer an

assortment of features that may not match the names. Despite offering a system dubbed Full Self-Driving, Tesla stated that its system did not autonomously handle all driving tasks. In the United Kingdom, a fully self-driving car is defined as a car so registered, rather than one that supports a specific feature set. The Association of British Insurers claimed that the usage of the word autonomous in marketing was dangerous because car ads make motorists think "autonomous" and "autopilot" imply that the driver can rely on the car to control itself, even though they do not.

Autonomous vehicle -- Part 4

2010s In a 2011 online survey of 2,006 US and UK consumers, 49% said they would be comfortable using a "driverless car". A 2012 survey of 17,400 vehicle owners found 37% who initially said they would be interested in purchasing a "fully autonomous car". However, that figure dropped to 20% if told the technology would cost US\$3,000 more. In a 2012 survey of about 1,000 German drivers, 22% had a positive attitude, 10% were undecided, 44% were skeptical and 24% were hostile. A 2013 survey of 1,500 consumers across 10 countries found 57% "stated they would be likely to ride in a car controlled entirely by technology that does not require a human driver", with Brazil, India and China the most willing to trust automated technology. In a 2014 US telephone survey, over three-quarters of licensed drivers said they would consider buying a self-driving car, rising to 86% if car insurance were cheaper. 31.7% said they would not continue to drive once an automated car was available. In 2015, a survey of 5,000 people from 109 countries reported that average respondents found manual driving the most enjoyable. 22% did not want to pay more money for autonomy. Respondents were found to be most concerned about hacking/misuse, and were also concerned about legal issues and safety. Finally, respondents from more developed countries were less comfortable with their vehicle sharing data. The survey reported consumer interest in purchasing an AC, stating that 37% of surveyed current owners were either "definitely" or "probably" interested. In 2016, a survey of 1,603 people in Germany that controlled for age, gender, and education reported that men felt less anxiety and more enthusiasm, whereas women showed the opposite. The difference was pronounced between young men and women and decreased with age. In a 2016 US survey of 1,584 people, "66 percent of respondents said they think autonomous cars are probably smarter than the average human driver". People were worried about safety and hacking risk. Nevertheless, only 13% of the interviewees saw no advantages in this new kind of cars. In a 2017 survey of 4,135 US adults found that many Americans anticipated significant impacts from various automation technologies including the widespread adoption of automated

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vehicles. In 2019, a test of a method of surveying opinions about self-driving cars, termed the Autonomous Vehicle Acceptance Model (AVAM), found that users were less accepting of high autonomy levels and displayed significantly lower intention to use highly autonomous vehicles. Additionally, partial autonomy (regardless of level) was perceived as requiring uniformly higher driver engagement (usage of hands, feet and eyes) than full autonomy.

Autonomous vehicle -- Part 5

Behavior prediction ACs must be able to predict the behavior of possibly moving vehicles, pedestrians, etc, in real time in order to proceed safely. The task becomes more challenging the further into the future the prediction extends, requiring rapid revisions to the estimate to cope with unpredicted behavior. One approach is to wholly recompute the position and trajectory of each object many times per second. Another is to cache the results of an earlier prediction for use in the next one to reduce computational complexity.

Autonomous vehicle -- Part 6

Notification In accordance with the proposed SAE J3134 "ADS Marker Lamps" specification, the US has standardized the use of turquoise lights to inform other drivers that a vehicle is driving autonomously. It will be used in the 2026 Mercedes-Benz EQS and S-Class sedans with Drive Pilot, an SAE Level 3 driving system. As of 2023, the turquoise light had not been standardized in China or by the UNECE.